



INITIAL SOLUTION PROPOSAL

PACK WISE AI - PACK PATROL

TEAM & PROJECT OVERVIEW

WHO ARE WE?

Project Title: Pack Wise

Team Name: Pack Patrol

University: President University

Team Members:

- Cristine Valentina
- Nasywa Chonifahtun F
- Shanty

WHAT PROBLEM ARE WE SOLVING?

Project Objectives:

- Build an AI advisor that recommends the optimal attachment type, quantity, and location.
- Balance product pose quality, drop-test performance, and customer experience.
- Use historical packaging outcomes to support performance evaluation and suggest simpler attachment designs that maintain retention quality.

OBSERVATION & PROBLEM UNDERSTANDING

What did we learn during the PTMI Visit?

OBSERVATION

During our visit to the PTMI assembly plant, we gained firsthand exposure to the doll packing process and the specific strap materials utilized.

IMPACT

- Brainstorming directly with our Mattel coach allowed us to instantly validate our AI ideas and ensure our solution fits real factory needs.
- We know what to solve: reducing plastic waste, cutting assembly time, and minimizing cost

ADDITIONAL INSIGHTS

The visit provided clear visibility into the project's scope, its limitations, and the specific data available for us to leverage. Crucially, we learned that Mattel possesses highly standardized drop-test protocols and meticulously tracked assembly time data for their production lines, making an AI-driven approach highly feasible.

SCOPE

Out-of-Scope (Project Limitations):

- Multi-item packaging: No processing of boxes with multiple dolls or complex accessories.
- Physical automation: No development of hardware or factory assembly robots.
- Product & box redesign: No changes to Barbie design or outer cardboard box shape.
- Live integration: The MVP is a standalone app, not linked to Mattel's factory systems.

PROPOSED SOLUTION

WHAT SOLUTION DO WE PROPOSE?

Solution Overview:

PackWise-AI is an AI-powered packaging optimization platform. It automatically scans products (like toys or figures) to determine the most secure, cost-effective, and eco-friendly way to package them, completely eliminating manual guesswork.

Why This Solution?

Traditional packaging design is slow, wastes materials, and relies on trial and error. PackWise-AI solves this by:

- Saving Money & Time: Removes unnecessary packaging straps, speeding up factory assembly and cutting material costs.
- Reducing Damage & Waste: Ensures products survive shipping safely while prioritizing sustainable materials to reduce plastic waste.

Key Features:

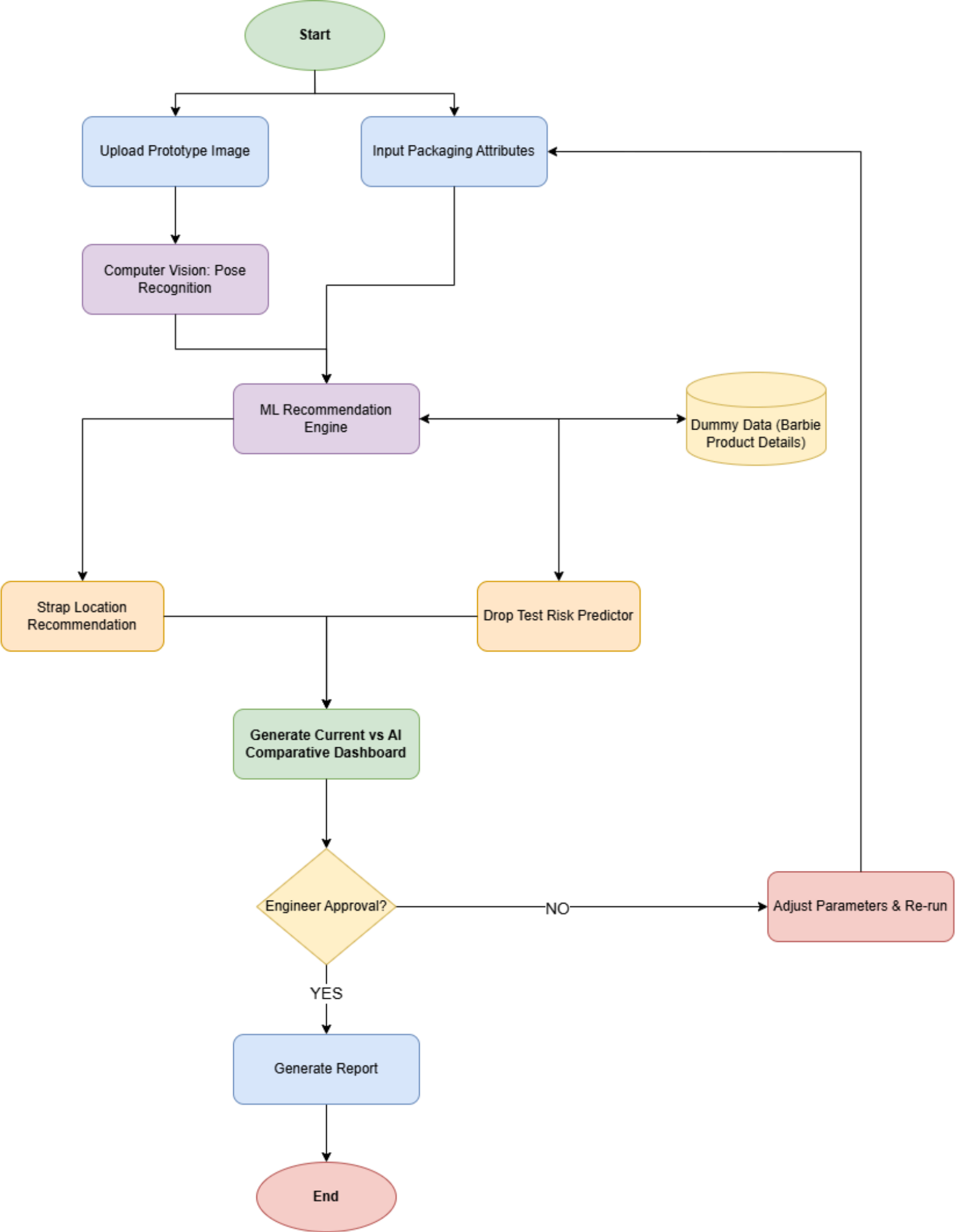
- AI Attachment Planner: Uses Computer Vision to scan a product's pose and instantly recommends exactly where straps or supports are needed.
- Risk & Sustainability Scoring: Predicts how well the packaging will hold up during shipping and tracks the environmental impact of the materials used.

SOLUTION HIGH LEVEL CONCEPT

FLOWCHART

The flowchart may not be fully visible due to slide size. Please refer to the Google Drive link below for the full-resolution version.

Pack Wise Flowchart



MVP PLAN

Technology Stack: React 19, Vite, TanStack Router, TypeScript, Tailwind CSS, Radix UI, Recharts, Supabase, Python, XGBoost, and YOLOv8.

What will we build during the Bootcamp?

Core MVP Features	Description
Adaptive AI Recommendation Engine	An intelligent system that recommends optimal attachment strategies and continuously learns from packaging engineers' feedback to improve future accuracy (Human-in-the-Loop).
Computer Vision & Skeleton Visualization	An input module that scans a product image, extracts its physical structure (skeleton), and provides a clear visual overlay of the exact recommended strap positions on the product.
SIOC Rule-Based Drop Test Predictor	Explain how this feature integrates with the rest of the MVP to form a cohesive prototype.A predictive module utilizing predefined engineering rules to assess whether the packaging design will pass drop-test standards specifically for Ships In Own Container (SIOC) requirements.

EXPECTED OUTCOME

EXPECTED BENEFITS

Operational Impact:

- Rapid Design: Replaces manual trials with instant visual AI.
- Fewer Physical Tests: Predicts SIOC compliance digitally.
- Standardized Quality: AI standardizes decisions via feedback loops.

Business Impact:

- Save Cost: Cuts plastic waste.
- Lower Risk: Prevents shipping damage & returns.
- Faster Launch: Speeds up packaging approvals.

QUESTIONS / ASSUMPTIONS

- Are there strict aesthetic guidelines or formulas for how different doll families must be posed? To maximize packaging efficiency and reduce straps, would it be possible to standardize most dolls into a 'neutral standing' pose?
- What are the exact cost variables for the different strap materials we observed during the plant visit?
- Are there strict 'must-strap' zones (like the neck or ankles) for all dolls, or is the AI allowed to completely reimagine the attachment locations to save material?